Spin-polarization control of photoelectrons using Poincaré fields

SUPPLEMENTAL MATERIAL

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I. VECTOR POTENTIALS OF THE MAGNETIC MULTIPOLE CLASS OF POINCARÉ FIELDS

In this section, we write the explicit form of the vector potentials for the magnetic multipole class of Poincaré fields used in the main text. To reiterate, in the far field, the generic form of the vector potential is given in Fourier- and configuration-space as [1, 2]:

$$\mathbf{A}_{jm}(\mathbf{k},t) = \Re\, C\, \mathbf{\Phi}_{jm}(\vartheta_{\hat{\mathbf{k}}}, \varphi_{\hat{\mathbf{k}}})\, \delta(|\mathbf{k}| - \omega)\, e^{-i\omega t} \tag{1}$$

$$\mathbf{A}_{jm}(\mathbf{r},t) = \Re\, C\, g_j(kr)\, \mathbf{\Phi}_{jm}(\vartheta, \varphi)\, e^{-i\omega t} \tag{2}$$

wherein $\Re$ denotes the real component of the entire quantity, $\hat{\mathbf{k}} = \mathbf{k}/\omega$ is the unit wavevector, and C is an arbitrary constant that will be used to introduce the normalized field amplitude

$$a_0 \equiv \frac{|e|E_0}{\omega m_e} = C g_j(\text{argmax})\, \mathbf{\Phi}_{jm}(\text{argmax}).$$

Here, "argmax" signifies the argument(s) of the maximum function value, and $g_j(z) = \sqrt{\frac{\pi}{2z}}\, J_{j+1/2}(z)$ are spherical Bessel functions of the first kind, defined recursively (Ref. [3], §10.1.10):

$$\begin{aligned} &g_n(z) \equiv f_n(z) \sin z + (-1)^{n+1} f_{-n-1}(z) \cos z, \\ &f_0(z) = z^{-1}, \quad f_1(z) = z^{-2}, \\ &f_{n-1}(z) + f_{n+1}(z) = (2n+1) z^{-1} f_n(z). \quad (n = 0, \pm1, \pm2, \ldots) \end{aligned} \tag{3}$$

$\mathbf{\Phi}_{jm}(\vartheta, \varphi) = \mathbf{r} \times \boldsymbol{\nabla} Y_{jm}(\vartheta, \varphi)$ are vector spherical harmonics, and (j, m) enumerate the total- and projected angular momentum mode numbers, $j \geq 1$ and $|m| < j - 1$. As discussed in the main text, $\mathbf{A}_{jm}(\mathbf{r}, t)$ is temporally modulated by a Gaussian-like quintic polynomial whose form is

$$p(\tau) = \begin{cases} 80\tau^3 - 240\tau^4 + 192\tau^5, & \text{for } 0 \leq \tau \leq 1/2 \\ 80(1-\tau)^3 - 240(1-\tau)^4 + 192(1-\tau)^5, & \text{for } 1/2 < \tau \leq 1 \end{cases} \tag{4}$$

where $\tau \in [0, 1]$ is the normalized pulse time. (Applying this factor is only valid in the far field, where the vector potential is initialized.)

A few remarks are in order about the vector potential expressions provided below. (i) The Bessel functions $g_j(kr)$ are expanded in terms of incoming (e^{-ikr}) and outgoing (e^{+ikr}) spherical waves. (ii) We denote $\mathbf{\Phi}'_{jm}(\vartheta, \varphi) \equiv \mathbf{\Phi}_{jm}(\vartheta, \varphi)/\mathbf{\Phi}_{jm}(\text{argmax})$, and note that for higher orders of j, the value of $\mathbf{\Phi}_{jm}(\text{argmax})$ has to be calculated numerically by maximizing $|\mathbf{\Phi}_{jm}| = (\mathbf{\Phi}_{jm} \cdot \mathbf{\Phi}^*_{jm})^{1/2}$ over the two-dimensional angle-space. To accomplish this, we used the Nelder–Mead optimization algorithm [4]. (iii) The spherical Bessel function maxima are also obtained numerically, and are provided in Table I. (iv) Modes for which $m \neq 0$ are obtained from their $m = 0$ counterparts by substituting the respective vector spherical harmonic, and so for those cases we provide just the $\mathbf{\Phi}_{j,m\neq0}(\vartheta, \varphi)$ form.

order j	max. value g_{j*}	arg. max. z_*
1	0.436182	2.081543
2	0.306792	3.342090
3	0.241746	4.514063
4	0.201582	5.646680
5	0.173970	6.756445

TABLE I. Maxima of the leading five spherical Bessel functions of the first kind, $g_j(z)$.

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$\underline{(j,m)=(1,0)}$

$$\mathbf{A}_{10}(\mathbf{r},t)=\Re\,\frac{a_0}{g_{1*}}\left[\left(\frac{-1}{2kr}-\frac{1}{2i(kr)^2}\right)e^{-ikr}+\left(\frac{-1}{2kr}+\frac{1}{2i(kr)^2}\right)e^{+ikr}\right]\mathbf{\Phi}'_{10}(\vartheta,\varphi)\,e^{-i\omega t} \tag{5}$$

with $\mathbf{\Phi}'_{10}(\vartheta,\varphi)=(-y/r,\;x/r,\;0)$

$\underline{(j,m)=(2,0)}$

$$\mathbf{A}_{20}(\mathbf{r},t)=\Re\,\frac{a_0}{g_{2*}}\frac{1}{2(kr)^3}\Big\{\big[3i-kr(ikr+3)\big]e^{-ikr}-\big[3i-kr(ikr-3)\big]e^{+ikr}\Big\}\mathbf{\Phi}'_{20}(\vartheta,\varphi)\,e^{-i\omega t} \tag{6}$$

with $\mathbf{\Phi}'_{20}(\vartheta,\varphi)=(2z/r^2)(-y,\;x,\;0)$

$\underline{(j,m)=(2,+1)}$

$$\begin{aligned}\mathbf{\Phi}'_{21}=\tfrac{1}{\sqrt{2}}e^{i\varphi}\Big\{&\big(i\cos^2\vartheta\cos\varphi+\cos 2\vartheta\sin\varphi\big)\hat{\boldsymbol{x}}\\&+\big(i\cos^2\vartheta\sin\varphi-\cos 2\vartheta\cos\varphi\big)\hat{\boldsymbol{y}}-i\cos\vartheta\sin\vartheta\,\hat{\boldsymbol{z}}\Big\}\end{aligned} \tag{7}$$

$\underline{(j,m)=(2,-1)}$

$$\begin{aligned}\mathbf{\Phi}'_{2,-1}=\tfrac{1}{\sqrt{2}}e^{-i\varphi}\Big\{&\big(i\cos^2\vartheta\cos\varphi-\cos 2\vartheta\sin\varphi\big)\hat{\boldsymbol{x}}\\&+\big(i\cos^2\vartheta\sin\varphi+\cos 2\vartheta\cos\varphi\big)\hat{\boldsymbol{y}}-i\cos\vartheta\sin\vartheta\,\hat{\boldsymbol{z}}\Big\}\end{aligned} \tag{8}$$

$\underline{(j,m)=(3,0)}$

$$\begin{aligned}\mathbf{A}_{30}(\mathbf{r},t)=\Re\,\frac{a_0}{g_{3*}}\Bigg\{&\left(\frac{1}{2kr}-\frac{3i}{(kr)^2}-\frac{15}{2(kr)^3}+\frac{15i}{2(kr)^4}\right)e^{-ikr}\\&+\left(\frac{1}{2kr}+\frac{3i}{(kr)^2}-\frac{15}{2(kr)^3}-\frac{15i}{2(kr)^4}\right)e^{+ikr}\Bigg\}\mathbf{\Phi}'_{30}(\vartheta,\varphi)\,e^{-i\omega t}\end{aligned} \tag{9}$$

with $\mathbf{\Phi}_{30}(\vartheta,\varphi)=\dfrac{3}{16}\sqrt{\dfrac{7}{\pi}}(\sin\vartheta+5\sin 3\vartheta)\cdot(\sin\varphi,\;-\cos\varphi,\;0)$

$$\mathbf{\Phi}_{30}(\text{argmax})\approx 1.5416594 \tag{10}$$

$\underline{(j,m)=(3,+1)}$

$$\begin{aligned}\mathbf{\Phi}_{31}(\vartheta,\varphi)=&\frac{-i}{16}\sqrt{\frac{21}{\pi}}\cos\vartheta\big[2-5e^{2i\varphi}+5(e^{2i\varphi}-2)\cos 2\vartheta\big]\hat{\boldsymbol{x}}\\&-\frac{1}{8}\sqrt{\frac{21}{\pi}}\cos\vartheta\big[-1+5\cos 2\vartheta-5e^{2i\varphi}\sin^2\vartheta\big]\hat{\boldsymbol{y}}\\&-\frac{i}{32}\sqrt{\frac{21}{\pi}}e^{i\varphi}\big(\sin\vartheta+5\sin 3\vartheta\big)\hat{\boldsymbol{z}}\end{aligned} \tag{11}$$

$\mathbf{\Phi}_{31}(\text{argmax})\approx 1.8281832$

$\underline{(j,m)=(3,-1)}$

$$\begin{aligned}\mathbf{\Phi}_{3,-1}(\vartheta,\varphi)=&\frac{1}{16}\sqrt{\frac{21}{\pi}}e^{-i\varphi}\cos\vartheta\big[i\cos\varphi(3+5\cos 2\vartheta)+\sin\varphi(7-15\cos 2\vartheta)\big]\hat{\boldsymbol{x}}\\&+\frac{1}{16}\sqrt{\frac{21}{\pi}}e^{-i\varphi}\cos\vartheta\big[\cos\varphi(-7+15\cos 2\vartheta)+i\sin\varphi(3+5\cos 2\vartheta)\big]\hat{\boldsymbol{y}}\\&-\frac{i}{32}\sqrt{\frac{21}{\pi}}e^{-i\varphi}(\sin\vartheta+5\sin 3\vartheta)\hat{\boldsymbol{z}}\end{aligned} \tag{12}$$

$\mathbf{\Phi}_{3,-1}(\text{argmax})\approx 1.8281832$

$\underline{(j,m) = (3,+2)}$

$$\begin{aligned}\mathbf{\Phi}_{32}(\vartheta,\varphi) = {} & \frac{1}{4}\sqrt{\frac{105}{2\pi}}e^{2i\varphi}\sin\vartheta\big[-2\cos^2\vartheta\big(i\cos\varphi+\sin\varphi\big)+\sin^2\vartheta\sin\varphi\big]\hat{\boldsymbol{x}} \\ & +\frac{1}{8}\sqrt{\frac{105}{2\pi}}e^{2i\varphi}\sin\vartheta\big[\cos\varphi+3\cos 2\vartheta\cos\varphi-4i\cos^2\vartheta\sin\varphi\big]\hat{\boldsymbol{y}} \\ & +\frac{i}{2}\sqrt{\frac{105}{2\pi}}e^{2i\varphi}\cos\vartheta\sin^2\vartheta\,\hat{\boldsymbol{z}} \\ \mathbf{\Phi}_{32}(\text{argmax}) \approx {} & 1.1317405\end{aligned} \tag{13}$$

$\underline{(j,m) = (3,-2)}$

$$\begin{aligned}\mathbf{\Phi}_{3,-2}(\vartheta,\varphi) = {} & \frac{1}{4}\sqrt{\frac{105}{2\pi}}e^{-2i\varphi}\sin\vartheta\big[2ie^{i\varphi}\cos^2\vartheta+\sin^2\vartheta\sin\varphi\big]\hat{\boldsymbol{x}} \\ & +\frac{1}{8}\sqrt{\frac{105}{2\pi}}e^{-2i\varphi}\sin\vartheta\big[\cos\varphi+3\cos 2\vartheta\cos\varphi+4i\cos^2\vartheta\sin\varphi\big]\hat{\boldsymbol{y}} \\ & -\frac{i}{2}\sqrt{\frac{105}{2\pi}}e^{-2i\varphi}\cos\vartheta\sin^2\vartheta\,\hat{\boldsymbol{z}} \\ \mathbf{\Phi}_{3,-2}(\text{argmax}) \approx {} & 1.1317405\end{aligned} \tag{14}$$

$\underline{(j,m) = (4,0)}$

$$\begin{aligned}\mathbf{A}_{40}(\mathbf{r},t) = {} & \Re\,\frac{a_0}{g_{4*}}\bigg\{\bigg(\frac{i}{2kr}+\frac{5}{(kr)^2}-\frac{45i}{2(kr)^3}-\frac{105}{2(kr)^4}+\frac{105i}{2(kr)^5}\bigg)e^{-ikr} \\ & +\bigg(\frac{-i}{2kr}+\frac{5}{(kr)^2}+\frac{45i}{2(kr)^3}-\frac{105}{2(kr)^4}-\frac{105i}{2(kr)^5}\bigg)e^{+ikr}\bigg\}\mathbf{\Phi}'_{40}(\vartheta,\varphi)\,e^{-i\omega t} \\ \text{with }\ \mathbf{\Phi}_{40}(\vartheta,\varphi) = {} & \frac{15}{16\sqrt{\pi}}\big[\sin 2\vartheta\sin\varphi(1+7\cos 2\vartheta)\hat{\boldsymbol{x}} \\ & -\tfrac{1}{2}\cos\varphi(2\sin 2\vartheta+7\sin 4\vartheta)\hat{\boldsymbol{y}}+0\hat{\boldsymbol{z}}\big] \\ \mathbf{\Phi}_{40}(\text{argmax}) \approx {} & 2.2342462\end{aligned} \tag{15}$$

$\underline{(j,m) = (4,+1)}$

$$\begin{aligned}\mathbf{\Phi}_{41}(\vartheta,\varphi) = {} & \frac{3}{32}\sqrt{\frac{5}{\pi}}\bigg\{\frac{i}{4}\big[9+20\cos 2\vartheta+35\cos 4\vartheta+12e^{2i\varphi}\sin^2\vartheta\big(5+7\cos 2\vartheta\big)\big]\hat{\boldsymbol{x}} \\ & -\frac{1}{4}\big[9+20\cos 2\vartheta+35\cos 4\vartheta-12e^{2i\varphi}\sin^2\vartheta\big(5+7\cos 2\vartheta\big)\big]\hat{\boldsymbol{y}} \\ & -ie^{i\varphi}\sin 2\vartheta\big(1+7\cos 2\vartheta\big)\hat{\boldsymbol{z}}\bigg\} \\ \mathbf{\Phi}_{41}(\text{argmax}) \approx {} & 2.6761862\end{aligned} \tag{16}$$

$\underline{(j,m) = (4,-1)}$

$$\begin{aligned}\mathbf{\Phi}_{4,-1}(\vartheta,\varphi) = {} & \frac{3i}{16}\sqrt{\frac{5}{\pi}}e^{-i\varphi}\big[\cos^2\vartheta\cos\varphi(1+7\cos 2\vartheta)+i\sin\varphi(\cos 2\vartheta+7\cos 4\vartheta)\big]\hat{\boldsymbol{x}} \\ & +\frac{3}{16}\sqrt{\frac{5}{\pi}}e^{-i\varphi}\big[i\cos^2\vartheta\sin\varphi(1+7\cos 2\vartheta)+\cos\varphi(\cos 2\vartheta+7\cos 4\vartheta)\big]\hat{\boldsymbol{y}} \\ & -\frac{3i}{32}\sqrt{\frac{5}{\pi}}e^{-i\varphi}\sin 2\vartheta\big[1+7\cos 2\vartheta\big]\hat{\boldsymbol{z}} \\ \mathbf{\Phi}_{4,-1}(\text{argmax}) \approx {} & 2.6761862\end{aligned} \tag{17}$$

$\underline{(j,m)=(4,+2)}$

$$\begin{aligned}\mathbf{\Phi}_{42}(\vartheta,\varphi)=\frac{3}{8}\sqrt{\frac{5}{2\pi}}e^{i\varphi}\Big\{&-\frac{i}{4}\big[3+21\cos 2\vartheta+14e^{2i\varphi}\sin^2\vartheta\big]\sin 2\vartheta\,\hat{\boldsymbol{x}}\\&+\frac{1}{4}\big[3+21\cos 2\vartheta-14e^{2i\varphi}\sin^2\vartheta\big]\sin 2\vartheta\,\hat{\boldsymbol{y}}\\&+ie^{i\varphi}\big[5+7\cos 2\vartheta\big]\sin^2\vartheta\,\hat{\boldsymbol{z}}\Big\}\\\mathbf{\Phi}_{42}(\text{argmax})\approx 1.6210298&\end{aligned}\tag{18}$$

$\underline{(j,m)=(4,-2)}$

$$\begin{aligned}\mathbf{\Phi}_{4,-2}(\vartheta,\varphi)=&\frac{3}{4}\sqrt{\frac{5}{2\pi}}e^{-2i\varphi}\sin\vartheta\cos\vartheta\big[i\cos\varphi(-1+7\cos^2\vartheta)+\sin\varphi(1-7\cos 2\vartheta)\big]\hat{\boldsymbol{x}}\\&+\frac{3}{4}\sqrt{\frac{5}{2\pi}}e^{-2i\varphi}\sin\vartheta\cos\vartheta\big[\cos\varphi(-1+7\cos 2\vartheta)+i\sin\varphi(-1+7\cos^2\vartheta)\big]\hat{\boldsymbol{y}}\\&-\frac{3i}{8}\sqrt{\frac{5}{2\pi}}e^{-2i\varphi}\sin^2\vartheta\big[5+7\cos 2\vartheta\big]\hat{\boldsymbol{z}}\\\mathbf{\Phi}_{4,-2}(\text{argmax})\approx&\ 1.6210298\end{aligned}\tag{19}$$

$\underline{(j,m)=(4,+3)}$

$$\begin{aligned}\mathbf{\Phi}_{43}(\vartheta,\varphi)=\frac{3}{8}\sqrt{\frac{35}{\pi}}e^{3i\varphi}\sin^2\vartheta\Big\{&\big[3i\cos^2\vartheta\cos\varphi+\sin\varphi+2\cos 2\vartheta\sin\varphi\big]\hat{\boldsymbol{x}}\\&-\big[\cos\varphi+2\cos 2\vartheta\cos\varphi-3i\cos^2\vartheta\sin\varphi\big]\hat{\boldsymbol{y}}\\&-3i\cos\vartheta\sin\vartheta\,\hat{\boldsymbol{z}}\Big\}\\\mathbf{\Phi}_{43}(\text{argmax})\approx 1.4953601&\end{aligned}\tag{20}$$

$\underline{(j,m)=(4,-3)}$

$$\begin{aligned}\mathbf{\Phi}_{4,-3}(\vartheta,\varphi)=&\frac{3}{8}\sqrt{\frac{35}{\pi}}e^{-3i\varphi}\sin^2\vartheta\big[3i\cos^2\vartheta\cos\varphi-(1+2\cos 2\vartheta)\sin\varphi\big]\hat{\boldsymbol{x}}\\&\frac{3}{8}\sqrt{\frac{35}{\pi}}e^{-3i\varphi}\sin^2\vartheta\big[3i\cos^2\vartheta\sin\varphi+(1+2\cos 2\vartheta)\cos\varphi\big]\hat{\boldsymbol{y}}\\&-\frac{9i}{8}\sqrt{\frac{35}{\pi}}e^{-3i\varphi}\sin^3\vartheta\cos\vartheta\hat{\boldsymbol{z}}\\\mathbf{\Phi}_{4,-3}(\text{argmax})\approx&\ 1.4953601\end{aligned}\tag{21}$$

$\underline{(j,m)=(5,0)}$

$$\begin{aligned}\mathbf{A}_{50}(\mathbf{r},t)=&\Re\frac{a_0}{g_{5*}}\bigg\{\bigg(\frac{-1}{2kr}+\frac{15i}{2(kr)^2}+\frac{105}{2(kr)^3}-\frac{220i}{(kr)^4}-\frac{945}{2(kr)^5}+\frac{945i}{2(kr)^6}\bigg)e^{-ikr}\\&+\bigg(\frac{-1}{2kr}-\frac{15i}{2(kr)^2}+\frac{105}{2(kr)^3}+\frac{220i}{(kr)^4}-\frac{945}{2(kr)^5}-\frac{945i}{2(kr)^6}\bigg)e^{+ikr}\bigg\}\mathbf{\Phi}'_{50}(\vartheta,\varphi)\,e^{-i\omega t}\\\text{with }\mathbf{\Phi}_{50}(\vartheta,\varphi)=&\frac{15}{256}\sqrt{\frac{11}{\pi}}\big(2\sin\vartheta+7\sin 3\vartheta+21\sin 5\vartheta\big)\sin\varphi\,\hat{\boldsymbol{x}}\\&-\frac{15}{256}\sqrt{\frac{11}{\pi}}\big(2\sin\vartheta+7\sin 3\vartheta+21\sin 5\vartheta\big)\cos\varphi\,\hat{\boldsymbol{y}}+0\hat{\boldsymbol{z}}\\\mathbf{\Phi}_{50}(\text{argmax})\approx&\ 3.0104177\end{aligned}\tag{22}$$

$\underline{(j,m)=(5,+1)}$

$$\mathbf{\Phi}_{51}(\vartheta,\varphi)=\frac{1}{16}\sqrt{\frac{165}{2\pi}}\Big\{\frac{i}{8}\cos\vartheta\big[29-28\cos 2\vartheta+63\cos 4\vartheta+56e^{2i\varphi}\sin^2\vartheta\big(1+3\cos 2\vartheta\big)\big]\hat{\boldsymbol{x}} \\ -\frac{1}{8}\cos\vartheta\big[29-28\cos 2\vartheta+63\cos 4\vartheta-56e^{2i\varphi}\sin^2\vartheta\big(1+3\cos 2\vartheta\big)\big]\hat{\boldsymbol{y}} \\ -ie^{i\varphi}\sin\vartheta\big(1-14\cos^2\vartheta+21\cos^4\vartheta\big)\hat{\boldsymbol{z}}\Big\} \tag{23}$$

$$\mathbf{\Phi}_{51}(\mathrm{argmax})\approx 3.6235732$$

$\underline{(j,m)=(5,-1)}$

$$\mathbf{\Phi}_{5,-1}(\vartheta,\varphi)=\frac{1}{16}\sqrt{\frac{165}{2\pi}}e^{-i\varphi}\Big\{ \\ \frac{1}{8}\cos\vartheta\big[i\cos\varphi(15+28\cos 2\vartheta+21\cos 4\vartheta)+\sin\varphi(-43+84\cos 2\vartheta-105\cos 4\vartheta)\big]\hat{\boldsymbol{x}} \\ \frac{1}{8}\cos\vartheta\big[\cos\varphi(43-84\cos 2\vartheta+105\cos 4\vartheta)+i\sin\varphi(15+28\cos 2\vartheta+21\cos 4\vartheta)\big]\hat{\boldsymbol{y}} \\ -i\sin\vartheta\big[1-14\cos^2\vartheta+21\cos^4\vartheta\big]\hat{\boldsymbol{z}}\Big\} \tag{24}$$

$$\mathbf{\Phi}_{5,-1}(\mathrm{argmax})\approx 3.6235732$$

II. HELICITY EIGENSTATES

The background material in this section follows closely that of Ref. [2], §23. As in the main text, we employ units for which $\hbar=c=1$, and the metric signature $(+,-,-,-)$. The standard representation of the Dirac matrices is used, with $\boldsymbol{\sigma}$ the Pauli matrices:

$$\gamma^0\equiv\beta=\begin{pmatrix}\mathbb{1} & 0\\ 0 & -\mathbb{1}\end{pmatrix},\quad \boldsymbol{\gamma}=\begin{pmatrix}0 & \boldsymbol{\sigma}\\ -\boldsymbol{\sigma} & 0\end{pmatrix},\quad \boldsymbol{\alpha}=\gamma^0\boldsymbol{\gamma},\quad\text{and}\quad \mathbf{\Sigma}=\begin{pmatrix}\boldsymbol{\sigma} & 0\\ 0 & \boldsymbol{\sigma}\end{pmatrix}.$$

A free particle plane-wave state with definite values of momentum $p^\mu=(\varepsilon,\mathbf{p})$ and energy $\varepsilon=\sqrt{|\mathbf{p}|^2+m_e^2}$ is written,

$$\psi_{\pm p}(x)=\frac{1}{\sqrt{2\varepsilon}}u_{\pm p}e^{\mp ipx} \tag{25}$$

wherein the upper (lower) signs are taken for an electron (positron). The bispinors $u_{\pm p}$ are given as

$$u_p=\begin{pmatrix}\sqrt{\varepsilon+m_e}\,w\\ \sqrt{\varepsilon-m_e}\,(\mathbf{n}\cdot\boldsymbol{\sigma})\,w\end{pmatrix},\quad u_{-p}=\begin{pmatrix}\sqrt{\varepsilon-m_e}\,(\mathbf{n}\cdot\boldsymbol{\sigma})\,w'\\ \sqrt{\varepsilon+m_e}\,w'\end{pmatrix}, \tag{26}$$

and they are normalized according to

$$\overline{u}_{\pm p}u_{\pm p}=\pm 2m_e,\quad \overline{u}_p\equiv u^\dagger\gamma^0 \tag{27}$$

with the two-component non-relativistic spinors w, w'.

Unlike the spin component along an arbitrary z-axis, the helicity λ of a particle—its spin component along the direction of momentum $\mathbf{n}\equiv\mathbf{p}/|\mathbf{p}|$—is a "good" quantum number, as the operator $\mathbf{n}\cdot\mathbf{\Sigma}$ commutes with the Hamiltonian $H=\boldsymbol{\alpha}\cdot\mathbf{p}+\beta m_e$. Helicity states are plane waves for which the spinor $w=w^{(\lambda)}(\mathbf{n})$ is an eigenfunction of the operator $\frac{1}{2}(\mathbf{n}\cdot\boldsymbol{\sigma})$,

$$\tfrac{1}{2}(\mathbf{n}\cdot\boldsymbol{\sigma})w^{(\lambda)}=\lambda w^{(\lambda)}. \tag{28}$$

These eigenspinors are given explicitly as

$$w^{(\lambda=+1/2)}=\begin{pmatrix}e^{-i\varphi_n/2}\cos(\vartheta_n/2)\\ e^{i\varphi_n/2}\sin(\vartheta_n/2)\end{pmatrix},\quad w^{(\lambda=-1/2)}=\begin{pmatrix}-e^{-i\varphi_n/2}\sin(\vartheta_n/2)\\ e^{i\varphi_n/2}\cos(\vartheta_n/2)\end{pmatrix}, \tag{29}$$

where (ϑ_n,φ_n) are the direction angles of $\mathbf{n}$.

A. Integrated Helicity Density

Let us consider a free electron in the $\lambda = +1/2$ helicity state. For convenience, we will suppose that it is traveling in the z direction $(\vartheta_n, \varphi_n = 0)$ with momentum p, which is always possible by choice of coordinates. Then $w \equiv w^{(\lambda=+1/2)} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ along with $(\mathbf{n} \cdot \boldsymbol{\sigma})w = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, and the state in this case is

$$\psi_p(x) = \frac{1}{\sqrt{2}} \begin{pmatrix} \sqrt{1 + m_e/\varepsilon} \\ 0 \\ \sqrt{1 - m_e/\varepsilon} \\ 0 \end{pmatrix} C_1\, G(\mathbf{x}) \exp[-i(\varepsilon t - pz)].$$

Here, we have introduced a Gaussian envelope $G(\mathbf{x}) = \exp(-x^2/\mu_x^2 - y^2/\mu_y^2 - z^2/\mu_z^2)$ to produce a localized wavepacket with (μ_x, μ_y, μ_z) the widths. This construction allows the expressions that follow to be evaluated more readily. While this state $\psi_p(x)$ is no longer an exact solution of Eq. (29), we will suppose that the $\mu_{x,y,z}$ are large compared to some characteristic size such as the Compton wavelength λ_c. The normalization factor C_1 is

$$\int d^3x\, \psi_p^\dagger \psi_p = 1 \;\Rightarrow\; |C_1| = 2^{3/4}\pi^{-3/4}(\mu_x\mu_y\mu_z)^{-1/2}.$$

The momentum-space wavefunction is

$$\begin{aligned} \tilde{\psi}_p(k) &\equiv \frac{1}{(2\pi)^{3/2}} \int d^3x\, \psi_p(x) \exp[-i(k_x x + k_y y + k_z z)] \\ &= \begin{pmatrix} \sqrt{1 + m_e/\varepsilon} \\ 0 \\ \sqrt{1 - m_e/\varepsilon} \\ 0 \end{pmatrix} C_2 \exp\left[-\tfrac{1}{4}\mu_x^2 k_x^2 - \tfrac{1}{4}\mu_y^2 k_y^2 - \tfrac{1}{4}\mu_z^2 (k_z - p)^2\right] \exp(-i\varepsilon t) \end{aligned}$$

wherein the momentum conjugate variable $\mathbf{k} = (k_x, k_y, k_z)$ is to be distinguished from the eigenvalue p of $\psi_p(x)$, and similarly $\mathbf{n_k} \equiv \mathbf{k}/|\mathbf{k}|$ with its direction angles (ϑ, φ) is different than the electron wave's chosen propagation direction, $\mathbf{n}_\parallel \hat{\mathbf{z}}$. (The wavepacket is centered at $(0, 0, p)$ in k-space, but there is a non-vanishing width in each of the momentum directions.) Note: $C_2 = 2^{-5/4}\pi^{-3/4}(\mu_x\mu_y\mu_z)^{1/2}$, and $\int d^3k\, \tilde{\psi}_p^\dagger \tilde{\psi}_p = 1$.

Now, the helicity density is given by the expectation value

$$\zeta(k) = \langle \tilde{\psi}_p(k) | (\mathbf{n_k} \cdot \tfrac{1}{2}\boldsymbol{\Sigma}) | \tilde{\psi}_p(k) \rangle \tag{30}$$

which here equals

$$\begin{aligned} \zeta(k) &= |C_2|^2 \exp\left[-\tfrac{1}{2}\mu_x^2 k_x^2 - \tfrac{1}{2}\mu_y^2 k_y^2 - \tfrac{1}{2}\mu_z^2 (k_z - p)^2\right] \times \\ &\quad \frac{1}{2}\left(\sqrt{1 + m_e/\varepsilon},\ 0,\ \sqrt{1 - m_e/\varepsilon},\ 0\right) \begin{pmatrix} \cos\vartheta & -i\sin\vartheta\, e^{i\varphi} & \cdot & \cdot \\ i\sin\vartheta\, e^{-i\varphi} & -\cos\vartheta & \cdot & \cdot \\ \cdot & \cdot & \cos\vartheta & -i\sin\vartheta\, e^{i\varphi} \\ \cdot & \cdot & i\sin\vartheta\, e^{-i\varphi} & -\cos\vartheta \end{pmatrix} \begin{pmatrix} \sqrt{1 + m_e/\varepsilon} \\ 0 \\ \sqrt{1 - m_e/\varepsilon} \\ 0 \end{pmatrix} \\ &= |C_2|^2 \cos\vartheta\, \exp\left[-\tfrac{1}{2}\mu_x^2 k_x^2 - \tfrac{1}{2}\mu_y^2 k_y^2 - \tfrac{1}{2}\mu_z^2 (k_z - p)^2\right]. \end{aligned}$$

Consider the integral of $\zeta(k)$ over all of momentum space, after making the usual transition to spherical coordinates $(k_x, k_y, k_z) = k\,(\sin\vartheta\cos\varphi,\ \sin\vartheta\sin\varphi,\ \cos\vartheta)$,

$$\begin{aligned} \int d^3k\, \zeta(k) = \int_0^\infty \int_0^\pi \int_0^{2\pi} dk\, d\vartheta\, d\varphi\, k^2 \sin\vartheta\cos\vartheta\, |C_2|^2 \exp\left(-\tfrac{1}{2}\mu_x^2 k^2 \sin^2\vartheta\cos^2\varphi\right) \times \\ \exp\left(-\tfrac{1}{2}\mu_y^2 k^2 \sin^2\vartheta \sin^2\varphi\right) \exp\left[-\tfrac{1}{2}\mu_z^2 (k\cos\vartheta - p)^2\right]. \end{aligned}$$

This integral, evaluated numerically with arbitrary (non-zero) values of $\mu_{x,y,z}$ and p, equals $+1/2$ in accordance with the chosen helicity eigenvalue. Similarly, had we taken the $\lambda = -1/2$ eigenstate for $\psi_p(x)$, then $\int d^3k\, \zeta(k) = -1/2$. $\square$

III. HELICITY-DENSITY PLOTS FOR NEON ION Ne^{9+} WITH $m < 0$ MODES

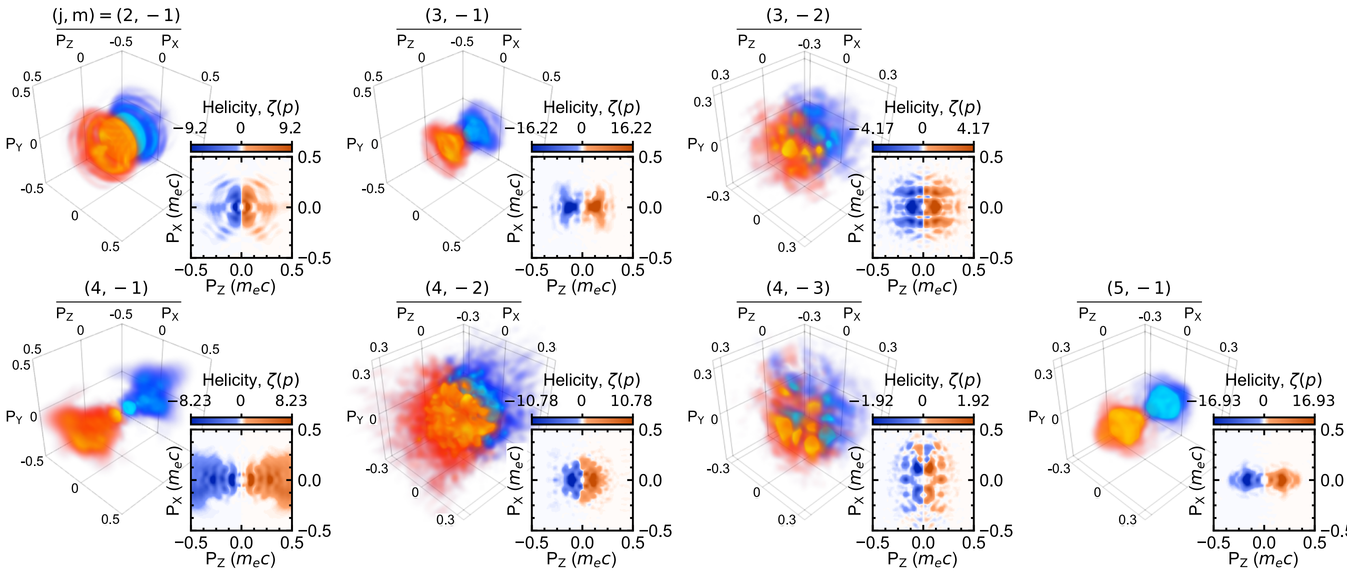

FIG. 1. Helicity density distributions $\zeta(\mathbf{p})$ of hydrogenic neon ion Ne^{9+} ($I_p/m_e \approx 2.67 \times 10^{-3}$) following irradiation by magnetic multipole fields of mode numbers $(j, m < 0)$ as labeled. Orange (blue) represents a positive (negative) helicity expectation value. Sub-panels: cross-sectional views for $p_y = 0$. As in the main text, the field amplitude is $a_0 = 10$ and the frequency is $\omega = 0.07\, m_e \approx 36$ keV ($\lambda \approx 0.0347$ nm, x-ray).

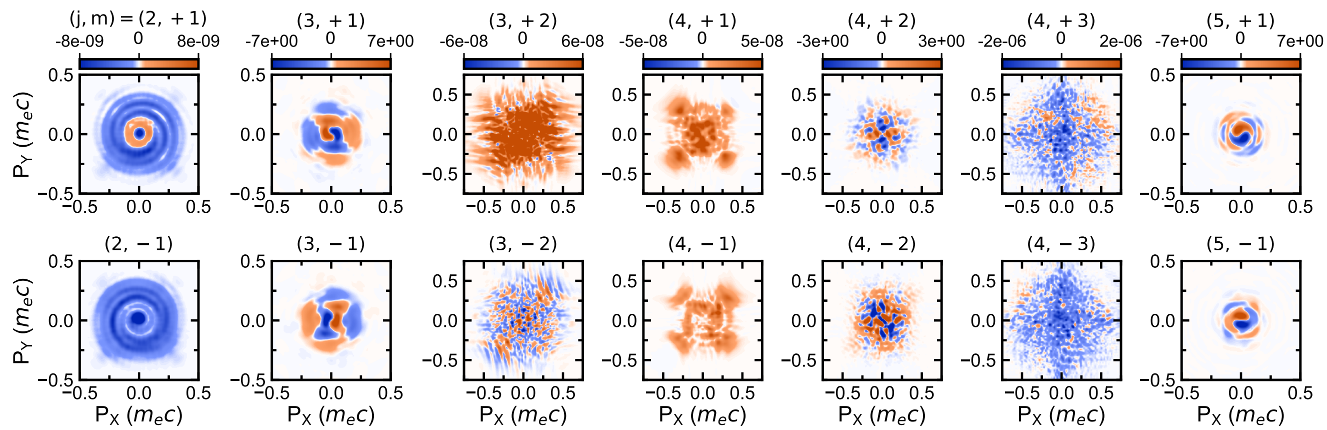

FIG. 2. The p_z-integrated helicity density distributions $\int dp_z\, \zeta(\mathbf{p})$ of hydrogenic neon ion Ne^{9+} for the cases of magnetic multipole irradiation with mode numbers $(j, m > 0)$ (top row) and $(j, m < 0)$ (bottom row), as labeled. Orange (blue) represents a positive (negative) helicity expectation value. The parameters are equivalent as in the main text and Fig. 1 above. For a given j value, the distributions for $\pm m$ employ the same colorbar scale.

IV. ADDITIONAL DATA FOR NEON ION Ne^{9+}

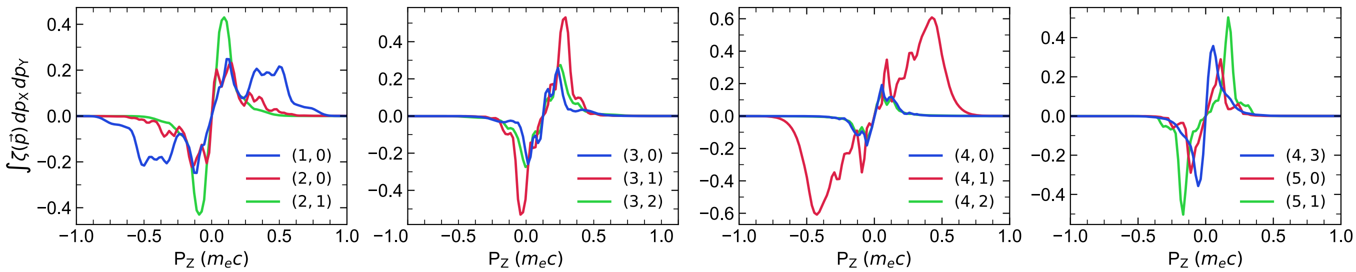

FIG. 3. The p_z-projected helicity density, $\int \zeta(\mathbf{p}) dp_x dp_y$, for each single-mode (j, m) field case of Fig. 2 in the main text.

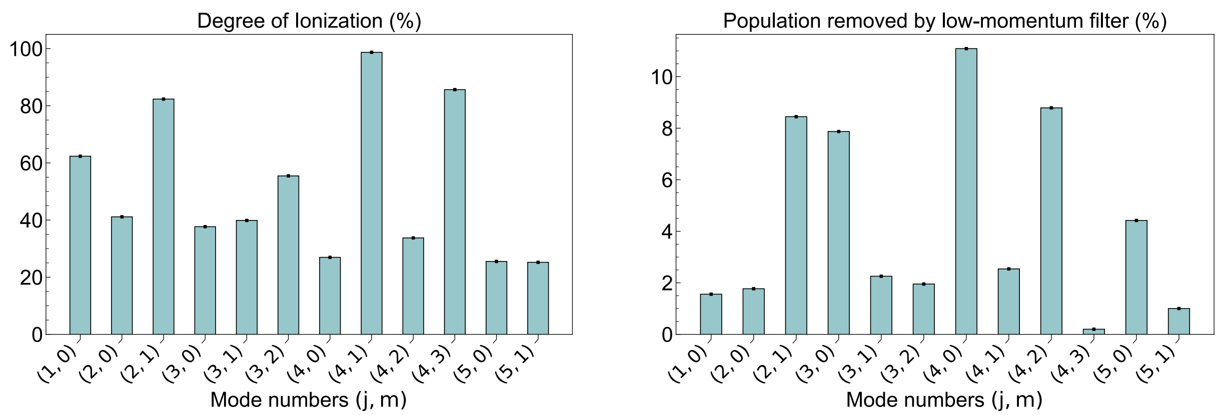

FIG. 4. (Single-mode ionization case). *Left:* Degree of ionization for each (j, m)-mode case of Fig. 2 in the main text. The bound population was removed by using a masking function of the form $\mathcal{M}_\mu(\mathbf{r}) = 1 - \exp(-|\mathbf{r}|^4/\mu^4)$, $\mu = 20\,\lambda_c$. *Right:* Percent of the population removed by the low-momentum filter $\mathcal{M}_\mu(\mathbf{p})$ with $\mu = 0.05\,m_e c$.

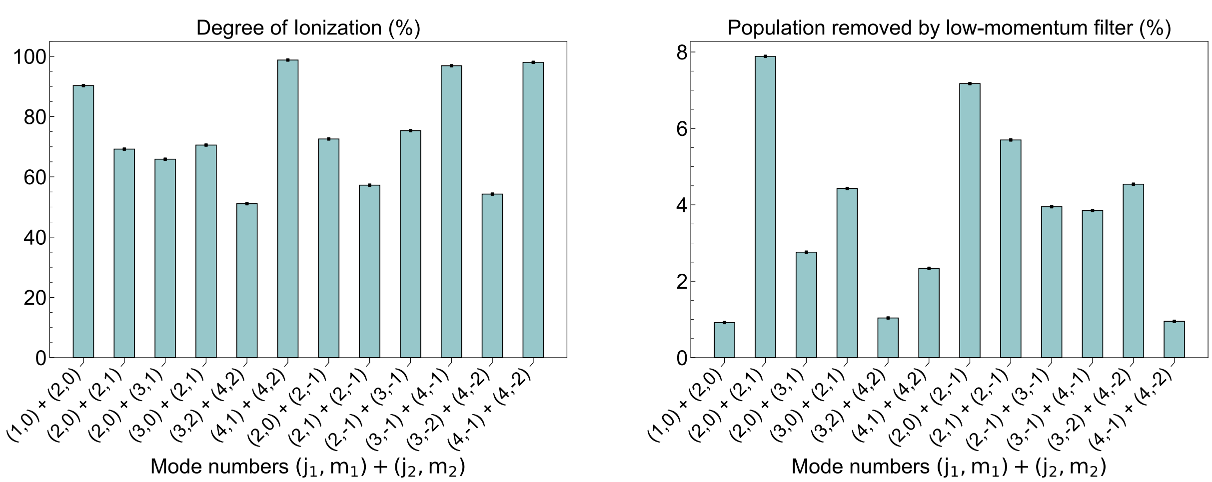

FIG. 5. (Multi-mode ionization case). Equivalent to Fig. 4 above, but for the multi-mode field cases presented in the main text.

V. PHOTOIONIZATION WITH CIRCULARLY-POLARIZED FIELDS

It is known from theory [5, 6] and experiment [7, 8] that circularly-polarized laser fields are capable of producing spin-polarized photoelectrons. In particular, the sign of the photoelectron helicity is dependent upon the polarization state of the incident photons, be it left- or right-circularly-polarized (LCP or RCP, respectively).

As a benchmark, we performed calculations of hydrogenic argon ion Ar^{17+} irradiated by plane, circularly-polarized fields. In this case, the peak amplitude is $a_0 = 50$, the central frequency is $\omega = 0.5\, m_e$, and the total pulse duration is 6 cycles. This simulation is heuristic. The intensity exceeds the Schwinger critical value, and the influence of other processes is not taken into account in a true quantum field-theoretic manner as required. It is intended only to illustrate how the net helicity changes sign between LCP and RCP field cases, and to draw a connection between the integrated helicity density and experimental measurements of the spin polarization.

In Fig. 6, one observes relativistic photoelectron momenta in the p_x-p_z helicity-density distributions (two leftmost panels). They are aligned in p_z as the field propagates in $+z$. Notice that the LCP case (panels 'a') predominantly yields photoelectrons of negative helicity, and conversely for the RCP case (panels 'b') with positive helicity. The integrated helicity expectation value $\int d^3p\, \zeta(\mathbf{p})$ is akin to what is measured experimentally, which is the net count of spin-up ($N_\uparrow$) and spin-down ($N_\downarrow$) photoelectrons, $S = (N_\uparrow - N_\downarrow)/(N_\uparrow + N_\downarrow)$ [7].

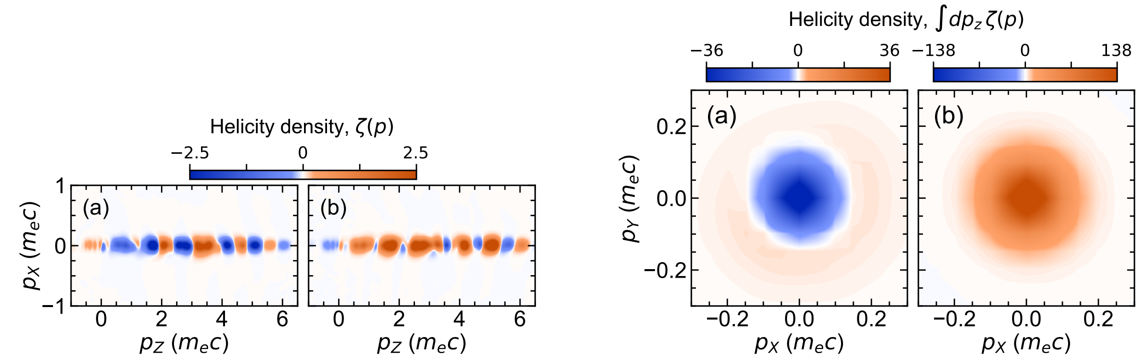

FIG. 6. Helicity density distributions $\zeta(\mathbf{p})$ of hydrogenic argon ion Ar^{17+} irradiated by (a) left- and (b) right-circularly-polarized fields. *Two leftmost panels:* Distributions in the x-z momentum plane, with $p_y = 0$. *Two rightmost panels:* Distributions integrated along the z momentum direction.

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